Learning Outcome:

After completing this hands-on exercise, you shall be able to:
• Load and explore a text dataset using Pandas by examining records, dimensions, column names, data types, missing values, and class distribution.
• Preprocess textual comments by cleaning text, separating input text from the target variable, encoding labels, and splitting the data into training and testing sets.
• Apply NLP preprocessing using tokenization and sequence padding to convert text into a format suitable for Deep Learning.
• Build, train and evaluate any Deep Learning Model (ANN, CNN, LSTM) using TensorFlow/Keras for multi-class toxic comment classification.
• Visualize model performance using accuracy/loss curves and a confusion matrix and test the trained model on new comments.

Problem Statement:

In this hands-on exercise, you will develop and evaluate a Deep Learning model for Toxic Comment Classification using the provided toxic_comments_dataset.csv dataset. The exercise will begin by loading and exploring the dataset using Pandas, including viewing sample records, checking its dimensions, column names, data types, missing values, and toxicity label distribution. You will then preprocess the comment text by cleaning the data, encoding the toxicity labels, and splitting the dataset into training and testing sets. After that, you will apply Natural Language Processing (NLP) techniques such as text tokenization and sequence padding to prepare the comments for Deep Learning.

Next, you will build and train any Deep Learning model (ANN, CNN, LSTM) using TensorFlow/Keras for classifying comments into their respective toxicity categories. Finally, you will evaluate the trained model using accuracy, loss, classification report, confusion matrix, and training/validation performance curves, and test the model on new comments to observe its predictions.

The purpose of this exercise is to provide practical experience with the complete NLP and Deep Learning workflow, from text preprocessing and tokenization to DL model development, training, prediction, and performance evaluation.

You can use Jupyter Notebook, Google Colab, VS Code, or any other Python programming environment/IDE to write and execute your code.

Use the following dataset in the repo named "toxic_comments_dataset".